DIY Ultrasonic Sensor report

Course: Sensors and Metrology
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1. Objective

The objective of this milestone is to build and operate a complete DIY ultrasonic sensor system using discrete electronic components and Arduino boards. The system integrates actuators, sensors, signal conditioning circuits, and microcontrollers to verify full system functionality. The work demonstrates practical understanding of ultrasonic sensing, signal amplification, and time-based detection.

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2. Hardware Design and Fabrication

2.1 Design Overview

The ultrasonic sensor system is designed as an alternative to commercial ultrasonic modules. It consists of a transmitter (TX) circuit for ultrasonic wave generation and a receiver (RX) circuit for echo detection and conditioning.

2.2 Materials and Tools

Category	Items
Active Components	Arduino Uno, Arduino Nano, LM324 Op-Amp, MAX232, BC557 Transistor
Passive Components	Resistors, Capacitors, Potentiometers
Sensors	Ultrasonic Transmitter (TX), Ultrasonic Receiver (RX)
Power	5V DC adaptor Supply
Tools	Soldering iron, jumper wires, breadboard, multimeter

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3. System Assembly

3.1 Assembly Process

Step	Description
1	Mount TX and RX transducers securely facing the target area
2	Assemble RX amplification circuit using LM324
3	Adjust potentiometers to obtain output signal > 3 Vpp
4	Connect MAX232 for signal shaping
5	Interface Arduino Uno with TX circuit
6	Interface Arduino Nano with RX output
7	Connect common ground between all modules

3.2 Critical Assembly Points

Point	Explanation
Interrupt Pin	Arduino Nano must use pin D2 for external interrupt
Grounding	All modules must share a common ground
Gain Adjustment	Incorrect potentiometer tuning may cause noise or missed echoes

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4. Full System Operation

Board	Function
Arduino Uno	Generates short trigger pulse (~12 µs) to activate TX
Arduino Nano	Detects echo pulse using external interrupt and serial output

4.1 Arduino Uno Code Explanation

1. Arduino Uno: Functional Overview

Item	Description
Main Function	Generates trigger pulse, receives echo pulse, calculates distance
Role in System	Master controller
Communication	Digital I/O with Arduino Nano
Measurement Principle	Time of Flight (ToF)

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2. Arduino Uno: Pin Configuration

Pin	Name	Direction	Function
D2	TRIG_OUT	Output	Sends trigger pulse to Arduino Nano
D3	ECHO_IN	Input	Receives echo pulse from Arduino Nano
GND	—	—	Common ground

3. Arduino Uno: Constant Definitions

Constant	Value	Purpose
SOUND_SPEED	343 m/s	Speed of sound in air at room temperature
Timeout	30 ms	Limits maximum measurable distance

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4. Arduino Uno: Setup Function

Step	Code Action	Explanation
1	Serial.begin(9600)	Initializes serial communication
2	Print startup message	Confirms program execution
3	pinMode(TRIG_OUT, OUTPUT)	Configures trigger pin
4	pinMode(ECHO_IN,INPUT)	Configures echo pin
5	digitalWrite(TRIG_OUT,LOW)	Ensures clean trigger start

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5. Arduino Uno: Main Loop Operation

5.1 Trigger Pulse Generation

Action	Description
Set TRIG_OUT HIGH	Starts trigger pulse
Delay 10 µs	Standard ultrasonic trigger duration
Set TRIG_OUT LOW	Ends trigger pulse

5.2 Echo Pulse Measurement

Step	Explanation
pulseIn(ECHO_IN, HIGH)	Measures echo pulse duration
Timeout = 30 ms	Prevents blocking if no echo

5.3 Distance Calculation

Parameter	Equation
Round-trip time	Measured by pulseIn()
Distance	d=(t×v)/2
Implemented As	(duration * 0.0343) / 2

Division by 2 accounts for forward and return travel.

5.4 Serial Output

Condition	Output
No echo	“No echo detected”
Echo detected	“Distance: XX.XX cm”

Arduino Nano Code Explanation

6. Arduino Nano: Functional Overview

Item	Description
Main Function	Generates ultrasonic burst and detects echo
Role in System	Slave / Sensor-side controller
Control Method	External interrupts
Signal Generation	Direct port manipulation

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7. Arduino Nano: Pin Configuration

Pin	Name	Function
D2	TRIG_IN (INT0)	Receives trigger from Uno
D3	ECHO_IN (INT1)	Receives conditioned RX signal
D4	TX Phase A	Ultrasonic burst
D6	TX Phase B	Ultrasonic burst (180° phase shift)
D5	Supply Control	Enables TX power
D10	ECHO_OUT	Sends echo pulse to Uno

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8. Arduino Nano: Global Flags

Variable	Type	Purpose
triggered	volatile bool	Indicates trigger received
echo_done	volatile bool	Indicates echo detected

Declared volatile because they are modified inside ISRs.

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9. Arduino Nano: Setup Function

Step	Code Action	Explanation
1	Serial.begin(9600)	Debug output
2	DDRD configuration	Sets TX and supply pins as outputs
3	DDRB configuration	Sets D10 as output
4	INPUT_PULLUP	Improves noise immunity
5	Attach INT0	Detects trigger pulse
6	Attach INT1	Detects echo signal

10. Ultrasonic Burst Generation (Nano)

Feature	Description
Frequency	≈ 40 kHz
Cycles	8 cycles
Phase Shift	D4 and D6 alternate
Timing	12 µs per half-cycle

Direct port manipulation ensures precise timing.

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11. Burst Transmission Sequence

Step	Explanation
Enable supply (D5 LOW)	Powers TX transducer
Alternate D4 & D6	Generates ultrasonic oscillation
Final half cycle	Ensures clean waveform
Power OFF	Prevents ringing

12. Echo Output Handling

Event	Action
Burst sent	D10 set HIGH
Echo detected	ISR sets echo_done = true
Echo complete	D10 set LOW

This pulse duration represents echo round-trip time.

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13. Interrupt Service Routines (ISRs)

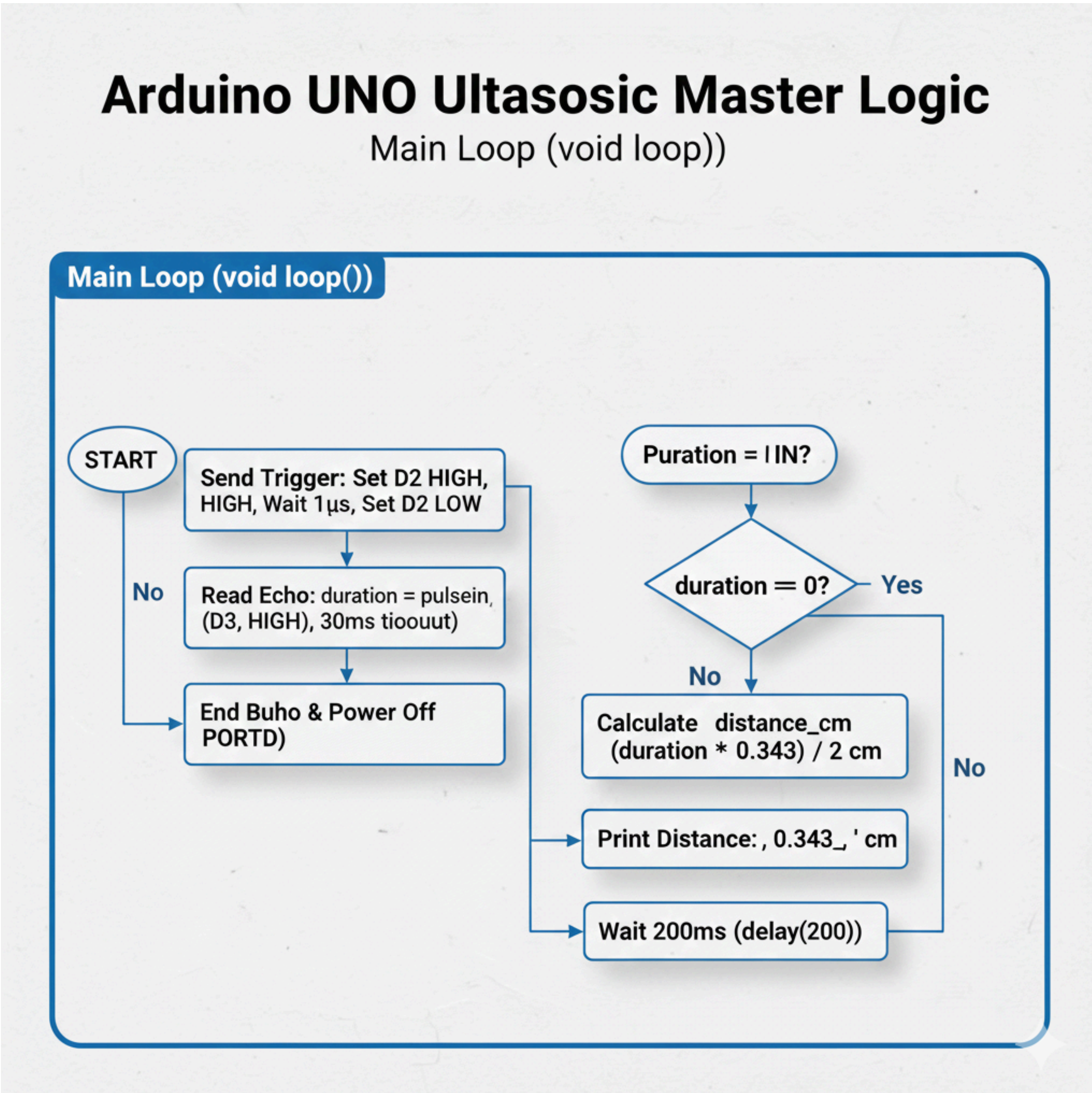
13.1 Trigger ISR

ISR	Function
triggerISR()	Sets triggered = true
Trigger Type	FALLING edge

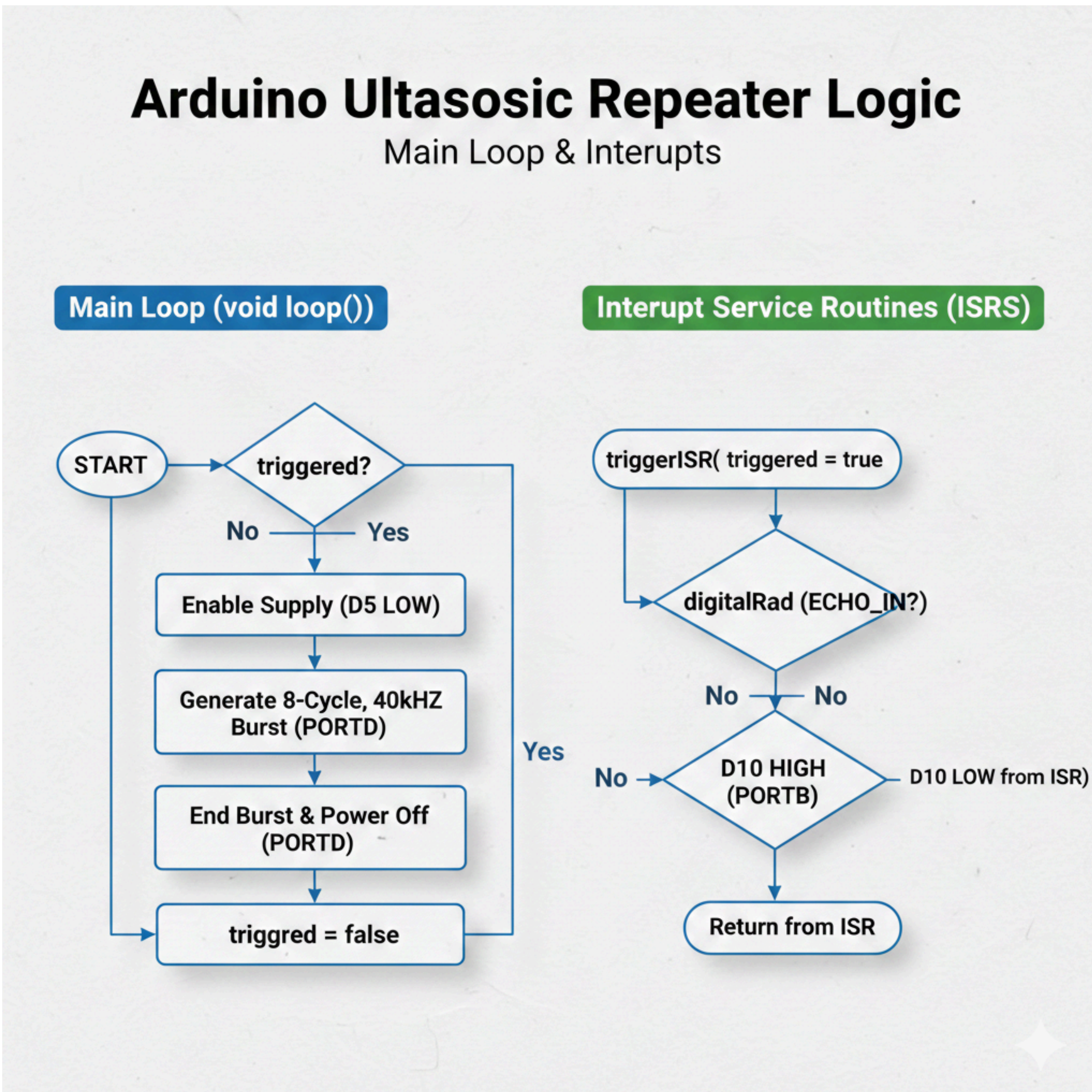
13.2 Echo ISR

ISR	Function
echoISR()	Sets echo_done = true
Trigger Type	CHANGE (rising/falling)

Arduino uno flowchart



Arduino nano flowchart



4.2 Operating Procedure 4.3 Expected Behavior

Step	Action
1	Power the system with 5V supply adapter
2	Open Serial Monitor on both boards
3	Arduino Uno sends periodic trigger pulses
4	TX emits ultrasonic wave
5	RX detects reflected wave
6	Arduino Nano confirms pulse reception

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5. Sensor Characteristics

5.1 Static Characteristics

Characteristic	Value / Description
Measurement Range	2 cm – 4 m (theoretical)
Sensitivity	High after multi-stage amplification
Resolution	Microsecond-level (timer dependent)
Accuracy	Affected by alignment, noise, and temperature, wires magnetic and electric field clash
Response Time	fast (µs range)

5.2 Justification of Sensor Selection

Requirement	Ultrasonic Sensor Suitability
Non-contact measurement	Yes
Medium-range sensing	Yes
Educational value	High (signal conditioning visibility)
Environmental tolerance	Works in low light and dusty conditions

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6. Results and Observations

Observation	Description
System Integration	All modules function together successfully
Echo Detection	Stable detection using interrupts
Noise Handling	Improved after gain tuning

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8. Conclusion

The DIY ultrasonic sensor system was successfully designed, assembled, and tested. The system demonstrates correct ultrasonic transmission and echo detection using discrete electronics and microcontrollers. This milestone confirms readiness for further development and distance measurement implementation.